Algebraic Formulation of the Collatz Operator in Jacobsthal Space: The $3J + J(0,1)$ Transformation

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Table 1. Collatz Orbit for Initial Value 7 in J-Spinor Representation

Step	Odd Node	Expanded J	Reduced J (Input)	3J+J(0,1)	k	Mapped J	Reduced J (Output)
1	7	J(4, 7)	J(4, 7)	J(12, 22)	1	J(6, 11)	J(6, 11)
2	11	J(6, 11)	J(6, 11)	J(18, 34)	1	J(9, 17)	J(4, 9)
3	17	J(9, 17)	J(4, 9)	J(12, 28)	2	J(3, 7)	J(2, 3)
4	13	J(7, 13)	J(2, 3)	J(6, 10)	1	J(3, 5)	J(0, 1)
5	5	J(3, 5)	J(0, 1)	J(0, 4)	2	J(0, 1)	J(0, 1)
6	1	J(0, 1)	J(0, 1)	J(0, 4)	2	J(0, 1)	J(0, 1)

The J-spinor transition process of the Collatz orbit from the initial node $b_0 = 7$ to the fixed point $J(0,1)$ (node 1). Each column represents the following algebraic operations:

- **Odd Node:** The current odd node in the standard Collatz orbit.
- **Expanded J:** The J sequence generated from the current odd node and its internal state $a = (b + 1)/2$ before J-space reduction.
- **Reduced J (Input):** The reduced spinor extracted as the representative element of the corresponding J equivalence class, serving as the input vector for the current step.
- **$3J + J(0,1)$:** The result of applying the affine transformation to the input spinor.
- **k:** The number of shifts applied (equivalent to division by 2^k in the standard Collatz operation).
- **Mapped J:** The normalized odd spinor obtained through the Shift & Cut normalization process without introducing fractional components.
- **Reduced J (Output):** The reduced spinor extracted from Mapped J, serving as the input for the next step.

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